VISUAL TREATMENT · V1.0

Apple-Style UI Animations

A motion, sound, and visual style bible
for InVideo creative agents.

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Reference document for AI-generated UI motion graphics

Use this to brief any UI animation request, in any context

CONTENTS

What's inside.

This document is a permanent reference. Attach it to any agent prompt where the goal is to produce Apple-feel UI motion. The agent should treat every section as binding.

01	The philosophy	Why Apple motion feels different
02	Motion principles	The 8 rules every animation follows
03	Easing & timing	Exact curves, durations, spring physics
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01 · PHILOSOPHY

Why Apple motion feels different.

Apple motion isn't decorative. It's confirmation. Every animation tells the user that their action registered and that the system is responding with intent. The motion is the feedback — not a flourish on top of the feedback.

> *"The animation isn't an effect. The animation IS the interaction."*

The four feelings

Every Apple-style UI motion should evoke at least one of these:

01

Weighted, not floaty

Elements have mass. They overshoot slightly, settle with a soft bounce. They don't slide linearly — they react to physics.

02

Confident, not eager

Motion is calm and decisive. Never twitchy. The element knows exactly where it's going and arrives without fuss.

03

Tactile, not graphic

Surfaces feel real — soft shadows, gentle highlights, subtle depth. Glass, not flat colour. Materials, not shapes.

04

Quiet, not loud

Sound is felt more than heard. A soft click, a tactile pop, a gentle whoosh. Never a fanfare.

The core test: if you removed the animation entirely and just cut to the final state, would the user feel the system was broken? If yes, the animation is doing its job. If no, you're over-animating.

02 · MOTION PRINCIPLES

The 8 rules.

These are non-negotiable. Every Apple-style UI animation must obey all eight.

1. Spring physics, not linear

Movement follows a damped spring curve. Elements accelerate from rest, overshoot the target by a tiny amount (3–8%), and settle. Never linear, never simple ease-in-out for the primary motion.

2. Scale + opacity, not just position

When something enters frame, it grows from ~85–92% scale to 100% AS it fades from 0 to 1 opacity. The two animations happen simultaneously, both completing on the same frame.

3. Stagger, don't parallelise

Multiple elements never enter at the exact same moment. Each element is offset by 60–120ms from the previous. This creates the cascading "deal of cards" feel that's signature Apple.

4. Anchor from the bottom or centre

Widgets and cards animate from their bottom edge or geometric centre — never the top-left corner. Scale and translate transformations originate from where the action makes physical sense.

5. Soft shadows that scale with motion

As an element rises into view, its shadow grows and softens. As it lands, the shadow tightens. The shadow is part of the animation, not a static asset.

6. Blur during transitions

Fast movement applies a subtle Gaussian blur (1.5–3px) during the most rapid part of the motion. This reads as "depth of field reacting to speed" rather than digital noise.

7. The bounce is small

Overshoot is 3–8% maximum. A bounce of 15%+ reads as cartoonish. Apple's signature bounce is so subtle most people don't consciously see it — they only feel it.

8. Exit faster than entry

Elements take ~500–700ms to enter, ~250–400ms to leave. Entrance announces presence; exit gets out of the way. Asymmetric timing is critical.

> *If your animation feels "fine" but not "Apple," the answer is almost always: smaller bounce, longer entry, faster exit, more stagger.*

03 · EASING & TIMING

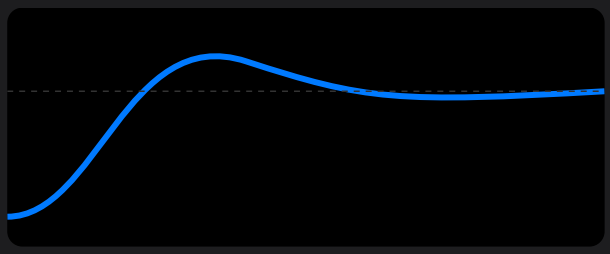
Exact curves and durations.

No guessing. These are the precise spring values and bezier curves to use. Plug them in directly.

The signature springs

SPRING — STANDARD ENTRY

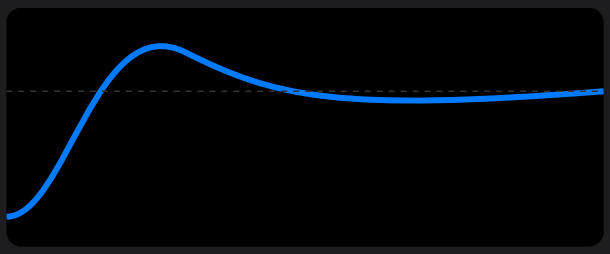
`stiffness: 180 · damping: 20 · mass: 1`



Default for most UI entries. ~600ms total. Soft 5% overshoot, settles cleanly. Use for cards, widgets, modals.

SPRING — SOFT POP

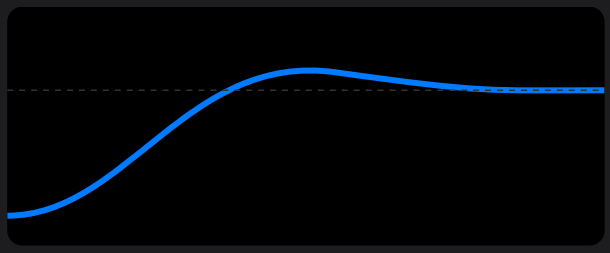
`stiffness: 260 · damping: 22 · mass: 1.1`



For small UI accents — toggles, badges, notification chips. ~450ms. Sharper attack, quick resolve.

SPRING — HEAVY SETTLE

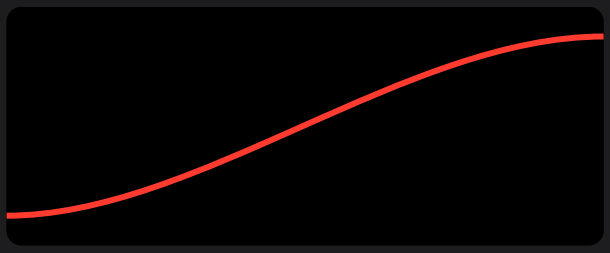
`stiffness: 140 · damping: 24 · mass: 1.3`



For large hero elements — full-screen widgets, big cards. ~800ms. Feels weighted and confident.

CUBIC BEZIER — EXIT

`cubic-bezier(0.32, 0, 0.67, 0)`



For exits. No bounce, just accelerate out. ~300ms. The element gets out of the way fast.

Duration cheat sheet

ELEMENT TYPE	ENTRY DURATION	EXIT DURATION	NOTES
Notification chip	450ms	250ms	Use Soft Pop spring

Widget / card	600ms	350ms	Standard Entry spring
Full-screen modal	800ms	400ms	Heavy Settle spring
Icon / app badge	500ms	250ms	Standard Entry, slightly faster
Text reveal	400ms per line	—	Stagger 80ms between lines
Progress fill	1200–1800ms	—	Ease-out cubic, no bounce
Dynamic Island morph	550ms	—	Custom spring, scale + corner radius simultaneously

Stagger timing

SCENARIO	STAGGER OFFSET	WHY
2–3 elements in a row	100ms	Reads as a group, but each gets its moment
4–6 elements (grid)	70ms	Tighter stagger keeps total duration under 1.2s
7+ elements (list reveal)	50ms	Approaches a cascade — feels alive
Multiple widget panels	120–150ms	Each panel feels weighted, deliberate

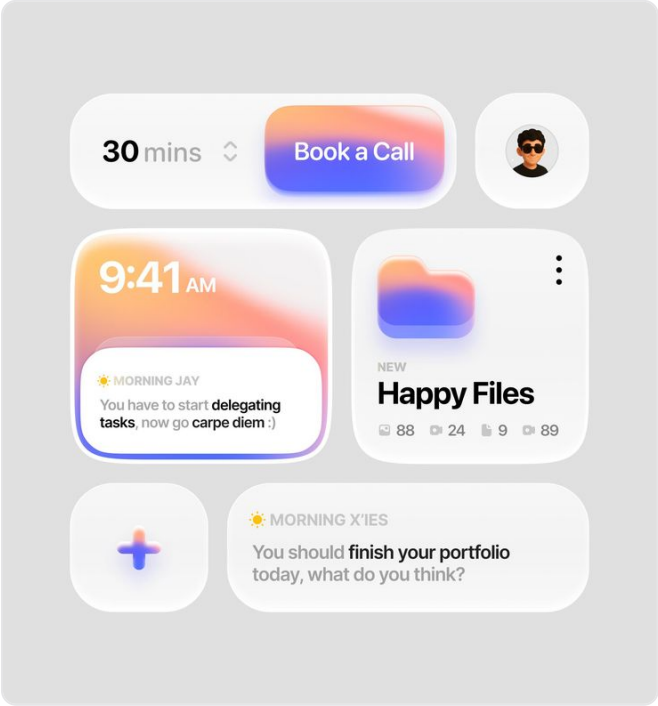
04 · THE VISUAL LIBRARY

What it looks like.

Every UI animation lives inside one of three visual systems. The agent should pick the right system based on context, then apply the correct material, lighting, and shadow language.

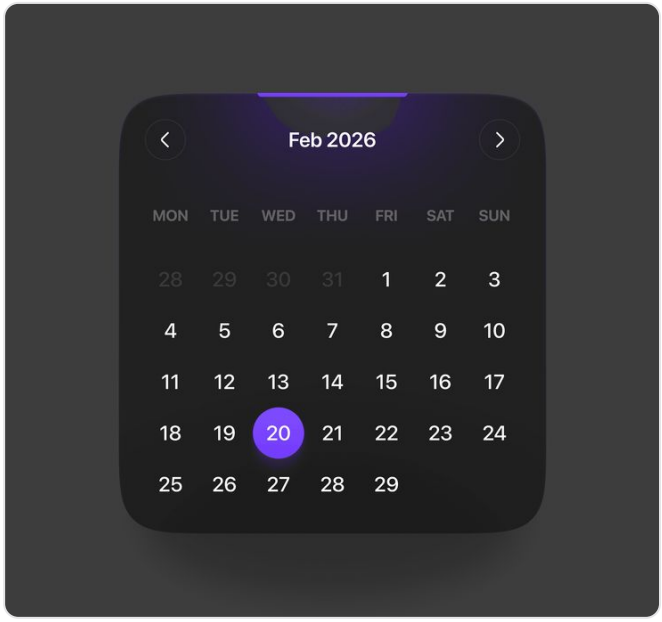
System A — Light mode (iOS / iPadOS)

Clean white or pale grey backgrounds. Soft drop shadows. Glassmorphism with high blur and low opacity. Vibrant accent colours pop against neutral surfaces.



Light system reference

Soft pill shapes, gradient pop accents, frosted glass cards. Drop shadows are diffuse and warm (slight blue undertone, never grey).

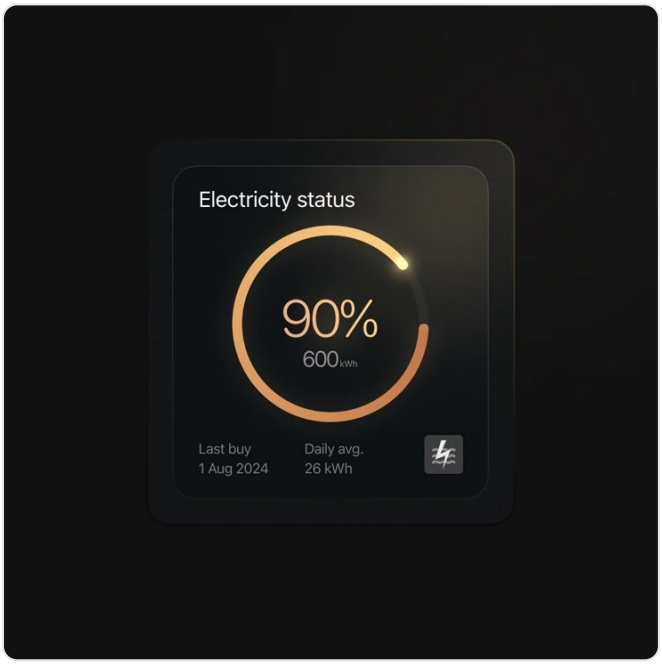


Component palette

Rounded pills, monochrome icons with single accent colour, dashed-line variants for empty/secondary states.

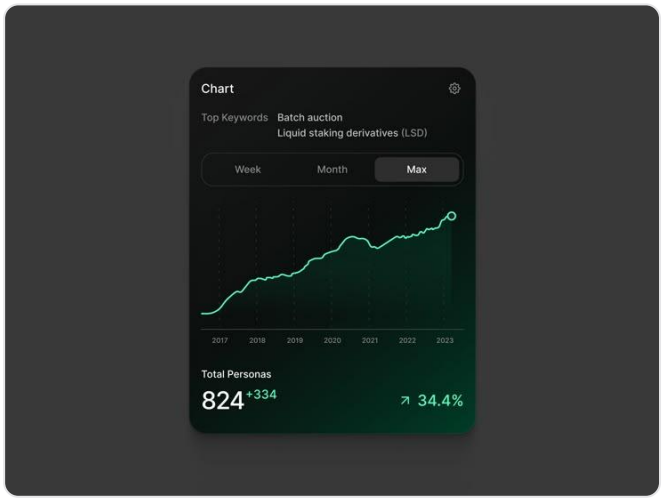
System B — Dark mode (System / Pro tools)

Deep black or near-black backgrounds (#0a0a0a or #000). Subtle inner highlights on cards. Glow effects on active elements. Accent colours feel like LED light against dark glass.



Dark surface card

Subtle gradient inside the card body (slightly lighter at top, falling off at bottom). Amber glow accent that bleeds into the surface.



Dark data visualisation

Green/teal accent with soft glow. Background grid is barely visible. Numbers are pure white, secondary text drops to 60% opacity.

System C — Hero / brand (gradient-driven)

Used for opening shots and hero moments. Vibrant gradients, glow halos, glassmorphic depth. Multiple light sources implied through colour temperature shifts.



Hero composition

Floating elements with individual depth. Each icon casts its own coloured glow into the background. Memoji-style 3D characters add character without dominating.



Backlit glow cards

Each card edge emits its own colour. The glow extends 30–50px beyond the card boundary. Background is pure black to maximise contrast.

Universal materials

Regardless of system, every surface uses one of these material treatments:

- **Frosted glass:** 20–30px backdrop blur, 70–85% opacity, subtle 1px inner border at 30% white
- **Soft plastic:** Solid fill with subtle gradient (3–5% lighter at top). 1px highlight at top edge.
- **Deep glass:** Used for dark mode hero. Heavy gradient, internal "glow from below" effect, edge highlights only.
- **Pure surface:** No texture, no gradient. Flat colour with a single soft shadow. Used for badges and pills.

05 · SOUND DESIGN

The audio fingerprint.

Sound is 50% of the Apple feel. Every animation has a paired sound — not a generic "click," but a specific frequency-range cue that matches the visual weight of the motion.

The sound library

Soft tactile pop `800Hz–1.2kHz · 80ms decay`	For element entries. A muted, dampened "tock" — like a finger on a stretched membrane. No metallic ring. Pair with Standard Entry spring.
Glass tick `2.5–4kHz · 60ms decay`	For toggles, switches, small confirmations. Higher frequency, very short. Should feel like tapping a fingernail on glass.
Whoosh `200Hz–6kHz sweep · 200–350ms`	For large element entries or exits. A frequency sweep — starts low, rises through the duration of the visual motion. Very low volume (-18dB).
Sub bump `40–80Hz · 120ms decay`	For hero moments — layered UNDER the tactile pop. Provides physical weight. Felt more than heard. Critical on phone speakers and AirPods.
Confirmation chime `C5 + G5 + C6 · 600ms decay`	For success states only. A soft, perfectly-tuned major triad. Reverb tail rolls off cleanly. Reserve for genuinely important moments — overuse kills it.
Friction drag `200–500Hz · sustained`	For sliders, scrubbers, scroll feedback. A continuous low-mid texture that pitches up with velocity. Stops instantly when motion stops.

Layering rules

- Hero moments stack 3 layers: **sub bump + tactile pop + light whoosh**. All hit within a 40ms window.
- Standard UI moments use 1 layer only. Resist the urge to stack.
- Every sound sits between -12dB and -24dB. Never louder. Sound design is felt, not heard.
- Sound peak hits at the moment of **visual settle** — not the start of the motion. The ear confirms what the eye just saw.

Sync rules

VISUAL EVENT	SOUND PEAK TIMING

Element enters frame	At first contact with overshoot peak
Element settles after bounce	At the moment of final rest
Element exits	At 30% through the exit motion
Toggle / state change	At the midpoint of the transition
Progress bar completes	At 100% fill, not before

Sources & libraries

- **Apple's own SF Symbols sounds** (lifted from iOS recordings) — the ground truth
- **Splice → "Apple UI" tag** — curated pack of iOS-feel one-shots
- **Native Instruments Komplete → Damage 2** for the sub bumps
- **FabFilter Pro-Q3** on every sound — high-pass everything below 30Hz, low-pass everything above 12kHz

06 · REFERENCE GALLERY

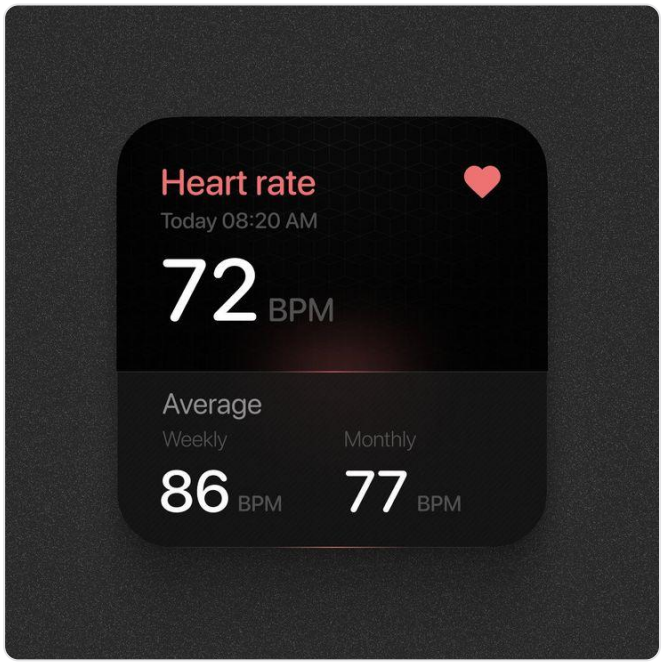
The look — annotated.

When briefing the agent, point to specific images from this gallery. "Make it feel like Reference 04" is more useful than describing the look in words.



Reference 01 — Balance card

Dark surface, pure white type hierarchy, single accent pill. Note the soft inner shadow at the card top. Animation: bottom-anchored entry, 600ms.



Reference 02 — Heart rate

Two-tone split layout. Numbers dominate, units drop to 60% opacity. Glow at the divider line. Animation: split reveal — top half enters first, bottom half 120ms later.



Reference 03 — Volume control

Discrete bar levels, not a continuous slider. Each bar animates in with 30ms stagger. Active state has soft outer glow.



Reference 04 — Navigation

Split composition with active indicator. The blue bar pulses subtly (1.5s loop, 95% to 100% opacity). Direction arrows are static.



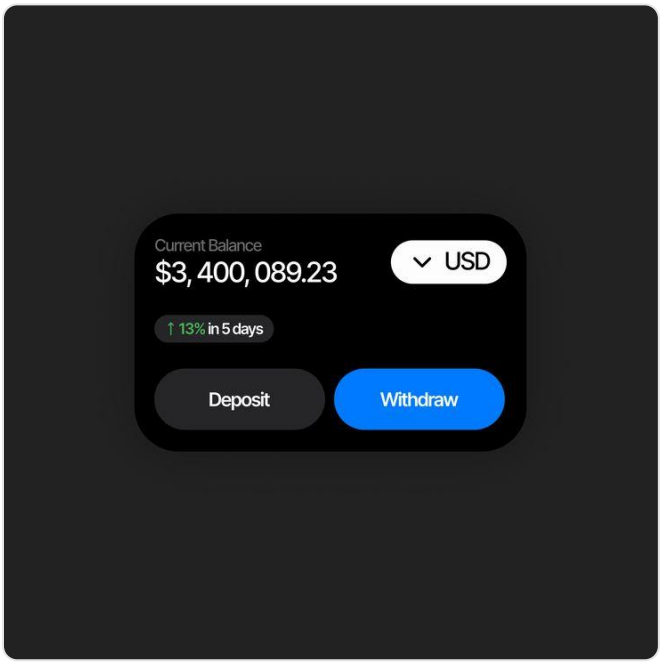
Reference 05 — Dynamic Island morph

The signature Apple motion. Pill expands with simultaneous corner radius animation. Use Heavy Settle spring. Sound: sub bump + soft pop.



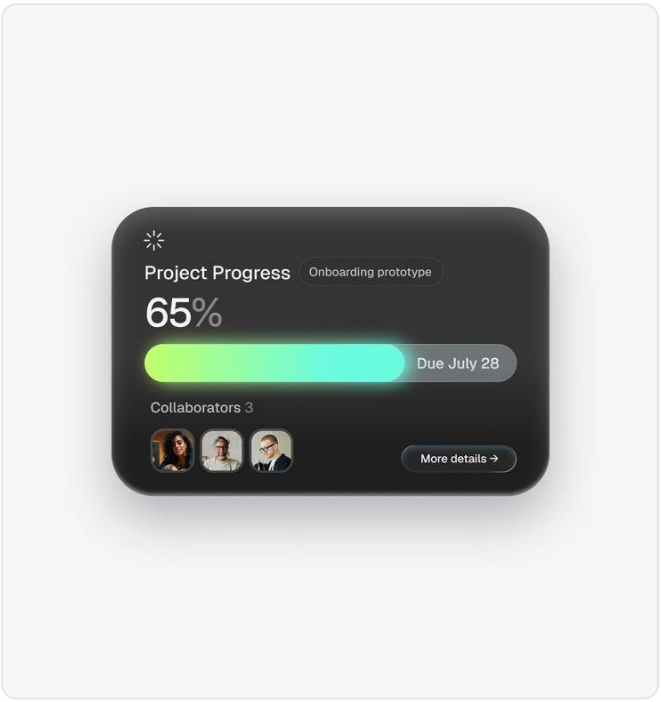
Reference 06 — Audio level

Live-feeling progress visualisation. Each bar has its own subtle delay. Active bars use saturated green; inactive sit at 30% opacity.



Reference 07 — Weather minimal

Maximum hierarchy contrast. Temperature is hero, location is supporting. Note the deliberate weight pairing: heavy number, regular label.



Reference 08 — Progress with glow

The fill bar GLOWS as it animates. Glow extends 8–12px beyond the bar. The glow is the animation's signature beat.



Reference 09 — Pill states

Each pill has its own colour identity AND its own glow. Backgrounds are translucent against pure black. Use for filter/state UI.



Reference 10 — Glass folder icon

Multi-layer glass with depth. The front face has higher opacity than the back. Inner highlight along the top edge. Used for hero icon moments.



Reference 11 — Hero composition. Multiple elements at varied depth, each with its own glow. Background is non-uniform — slight gradient washes from the implied light sources. This is the "money shot" treatment.

After Effects spec.

Exact expressions, plugin recommendations, and keyframe values. Copy and paste these directly.

Required plugins

- **Motion v4** (mt.mograph) — for proper spring physics and bounce
- **Flow** (Aescripts) — for easing curve presets and saving the curves from this doc
- **Overlord** — Figma → AE pipeline for getting clean vector assets
- **Deep Glow** — for the signature glow on dark-mode references
- **Pastiche** (optional) — for animated grid reveals

The signature spring expression

Apply to any property (Position, Scale, Rotation)

```
n = 0;
if (numKeys > 0){
  n = nearestKey(time).index;
  if (key(n).time > time) n--;
}
if (n == 0){ t = 0; } else { t = time - key(n).time; }

if (n > 0 && t < 1){
  v = velocityAtTime(key(n).time - thisComp.frameDuration/10);
  amp = 0.05; // 5% overshoot
  freq = 2.0;
  decay = 6.0;
  value + v*amp*Math.sin(freq*t*2*Math.PI)/Math.exp(decay*t);
} else { value; }
```

Adjust `amp` for overshoot amount, `freq` for bounce count, `decay` for how quickly it settles. The values above match the Standard Entry spring from Section 03.

The "enter from below" preset

PROPERTY	EASING

	FRAME 0	FRAME ~18 (600MS @ 30FPS)	
Position Y	+40px	0px	Spring expression above
Scale	88, 88%	100, 100%	Spring expression above
Opacity	0%	100%	Ease-out cubic, completes at frame 12
Drop Shadow Distance	5px	15px	Linked to Position Y
Drop Shadow Softness	8px	25px	Linked to Position Y

Anchor point matters. Before applying this preset, move the anchor point to the bottom-centre of the layer (Pan Behind tool + hold Cmd). Otherwise scale will distort the position.

Stagger expression for groups

```
// Apply to Time Remap or property on each layer
delay = (index - 1) * 0.08; // 80ms stagger
valueAtTime(time - delay);
```

Place all elements as siblings, name them sequentially. The expression auto-staggers based on layer index. Change `0.08` to adjust offset (see Section 03 stagger chart).

Glow setup (dark mode)

- Duplicate the layer. Set blending mode of the copy to **Add**.
- Apply Deep Glow: Radius 25, Brightness 1.2, Threshold 0.4
- Pre-comp the result. Animate Pre-comp opacity from 0 → 100% in sync with the main animation.
- For the "edge glow" look (Reference 09), mask the duplicate to a 2px outline only.

Fusion spec.

For when the UI animation lives inside a DaVinci Resolve project — common for sponsored content and longer-form video.

Node tree for a standard UI animation

```
Background ──┐
             ├──> Merge ──> Glow ──> Blur (animated) ──> ColorCorrector ──> MediaOut
UI Element ──┘
  │
  ├──> Transform (Position, Scale animated via spline)
  ├──> DropShadow (Distance + Softness animated)
  └──> Brightness/Contrast (Opacity via Gain)
```

The Fusion spring — via Spline Editor

Fusion doesn't have native spring physics. Build them manually in the Spline Editor:

1. Set keyframes at frame 0 (start state) and frame 18 (target state).
2. Add a third keyframe at frame 14 that EXCEEDS the target by 5% (the overshoot peak).
3. In the Spline Editor, set the curve tension on the middle keyframe to **0.0** (linear in and out).
4. Set the start keyframe to **Smooth** with bias +0.3 (slow start, fast finish).
5. Set the end keyframe to **Smooth** with bias -0.2 (settle into rest).

Fusion-specific gotchas

> **Anchor points work differently in Fusion.** Use the Transform node's Pivot parameter to set the anchor before keyframing. Default is centre — change to (0.5, 0.1) for bottom-anchored scaling.

Glow setup in Fusion

- Use **Soft Glow** node, NOT regular Glow — softer falloff, more Apple-feel
- Glow Size: 25–40 for cards, 8–15 for pills
- Glow Color: sample directly from the element's accent colour, then push saturation +20%

— Threshold: 0.4–0.55 — too low and the entire UI glows, too high and only the brightest pixels do

— Blend: 50–70% — never 100%

Resolve colour grade for UI animations

WHEEL	ADJUSTMENT	WHY
Lift	-0.02 (slight crush)	Deepens the blacks on dark-mode UI
Gamma	+0.05 R, +0.02 B	Subtle warm tint — matches iOS rendering
Gain	+0.08 highlights only	Brightens the accent colours without blowing out
Saturation	1.1 (10% boost)	iOS UI is more saturated than print-accurate

09 · ANTI-PATTERNS

What to never do.

If the agent produces any of these, reject the output immediately. These are the tells of generic motion graphics — the opposite of what we want.

Motion anti-patterns

- **Linear ease.** Never. Anything that moves at constant velocity is wrong.
- **Excessive bounce.** If the overshoot is visible without slowing the footage down, it's too much.
- **Everything entering at once.** Stagger is mandatory for any group of 2+ elements.
- **Top-left anchor scaling.** Elements that grow from their corner read as web ads, not iOS.
- **Faster entry than exit.** Apple is the opposite — slow entry, fast exit.
- **Same easing on every property.** Position, scale, and opacity should each have their own curve.
- **Rotation as a primary motion.** Apple UI almost never rotates. Don't add it for "interest."

Visual anti-patterns

- **Hard drop shadows.** Apple shadows are diffuse and large. A 2px hard shadow is After Effects-default — kill it.
- **Pure black (#000) text on white.** Use #1d1d1f. The slight warmth is the difference.
- **Saturated primary colours.** iOS system blue is #007AFF — not #0000FF. Always desaturate by ~10–15%.
- **Symmetrical glows.** Glow should fall off naturally based on the light source direction implied by the composition.
- **Flat backgrounds.** Even a "white background" has a subtle radial gradient. Pure flat reads as cheap.
- **Mixing material systems.** A frosted glass card sitting on a flat plastic background looks broken. Commit to one material world per shot.
- **Generic SF Pro substitutes.** If SF Pro isn't available, use Inter — never Helvetica, never Roboto, never (god forbid) Arial.

Sound anti-patterns

- **Cinematic risers and impacts.** Trailer sound design is the opposite of Apple sound design.
- **Sound on every micro-interaction.** Most UI moments are silent. Sound is for confirmation, not decoration.
- **Sound that peaks BEFORE the visual lands.** Sync the peak to the moment of settle, not the start of motion.

— **Long reverb tails on tactile sounds.** Pops and clicks should be dry. Reverb is reserved for confirmation chimes.

— **Sound louder than -12dB.** If anyone consciously notices the sound, it's too loud.

> *"If it looks like a stock motion graphics template, the agent has failed. Apple UI motion is invisible — it doesn't announce itself."*

10 · AGENT BRIEF TEMPLATE

How to invoke this style.

When briefing the InVideo agent, use this template. It forces the agent to consult every section of this document before producing output.

Standard brief format

Paste this template, fill in the bracketed sections, and attach this PDF as reference:

```
Produce an Apple-style UI animation following the attached
style bible (Sam John · UI Animation Style Bible v1.0).

ELEMENT TYPE: [widget / pill / card / icon / modal / chart]
VISUAL SYSTEM: [Light / Dark / Hero gradient]
CLOSEST REFERENCE: [Reference 01–11 from gallery]
ENTRY MOTION: [from below / from centre / scale-in / morph]
SPRING: [Standard Entry / Soft Pop / Heavy Settle]
SOUND: [tactile pop / glass tick / whoosh / sub bump / chime]
CONTEXT: [full-screen hero / overlay on footage / lower third]
DURATION: [total animation length in ms]
EXTRA: [glow / blur / stagger details / specific copy]

Reject any output that breaks Section 09 (Anti-patterns).
```

Example brief — completed

"Notification chip dropping in over my B-roll"

```
ELEMENT TYPE: notification chip
VISUAL SYSTEM: Dark (deep glass)
CLOSEST REFERENCE: Reference 05 (Dynamic Island morph)
ENTRY MOTION: from above, scale-in with corner-radius morph
SPRING: Soft Pop (260/22/1.1)
SOUND: sub bump (40Hz, -20dB) + tactile pop (1kHz, -16dB)
CONTEXT: overlay on footage, top-centre, 30% from top
DURATION: 450ms in, holds 2s, 250ms out
```

```
EXTRA: subtle 2px white inner border at 30% opacity,
       chip content reveals 100ms after pill settles
```

Quality checklist

Before accepting any agent output, verify:

- Spring physics applied (not linear, not basic ease)
- Scale + opacity animate together on entry
- Stagger between elements is at least 50ms
- Anchor point is bottom or centre (not corner)
- Shadow scales with motion
- Overshoot is between 3–8%
- Entry duration is longer than exit duration
- Sound peaks AT visual settle (not before)
- Sound sits between -12dB and -24dB
- Colours match the chosen visual system
- Typography uses SF Pro (or Inter as fallback)
- No items from Section 09 (Anti-patterns) are present

When the agent gets it wrong

The most common failure modes and how to correct them:

SYMPTOM	FIX
"Feels generic / motion graphic-y"	Reduce bounce, increase entry duration, add stagger
"Feels twitchy / nervous"	Reduce spring stiffness, increase damping
"Looks cheap"	Add soft glow, refine shadow softness, check material treatment
"Sound is annoying"	Drop volume 6dB, remove a layer, check sync timing
"Doesn't feel Apple"	Almost always: smaller bounce, more stagger, softer shadows

CLOSING

One last thing.

This document is a starting point, not a constraint. Apple's own design system evolves — and so should this. Update it as you discover new motion patterns, refine sound choices, and develop your own variations on the language.

> *"The best motion design feels inevitable. Like the element could have moved no other way. That's the target."*

Every UI animation you brief from now on should be able to point to a specific page of this document. If you can't, the request isn't specific enough yet — refine the brief before sending it to the agent.

Sam John · UI Animation Style Bible

Version 1.0 · Compiled May 2026

For use with InVideo creative agents and personal production workflows.